ANATOMY OF AI AGENTS 2026

www.arianprabowo.com · 1. Silicon → 10. Swarms × 4 Organisms

① COLUMNS (1 → 10): 1. Silicon → 2. Python → 3. Data → 4. Pre-Train → 5. SFT → 6. Post-Train → 7. Test-Time → 8. Context → 9. Harness → 10. Swarms
② ROWS (4 ORGANISMS): Core LLM Spine + Separated RAG, Multimodal & Embodied Organisms
③ INSIDE EVERY CELL: Flagship Pill (H2) + Checkpoints (H3) + Core Stack & Red ⚠ ANTI-PATTERNS Box (H4)

*Synthesized from Haggai Roitman's 'The Hitchhiker's Guide to Agentic AI: From Foundations to Systems' (arXiv:2606.24937, v1.4, June 2026) + 2026 systems tooling · Companion to AI Skill Tree 2026 & AI Ladder Evolution by Arian Prabowo · Updated: 2026-11-08 02:25:58 AEDT (UTC+11:00)

ANATOMY

1 → 10 Stages

LLM · RAG · Multimodal · Embodied (+ ⚠ Traps)

1. Silicon

Physical Compute Silicon, GPU Clusters, HBM3e Memory & Interconnect Fabrics

- Unit: FLOPs, HBM Bytes & Warps
- Guide: Ch. 1.6, 2.3, 11.10-14, 30.5

2. Python

Python Runtimes, Package Managers, Virtual Environments & Tensor Compilers

- Unit: Wheels, Tensors & Paged KV
- Guide: Ch. 1.24-25, 11.1-9, 21.5, 22.1

3. Data

Web & Code Corpora, Subword Tokenization & Curation Pipelines

- Unit: Corpora, Subwords & Shards
- Guide: Ch. 1.2, 3.1, 3.3-3.5, 4.1, 30.2

4. Pre-Train

Foundation Model Architectures, Sparse MoE Routing & Self-Supervised Pre-Training

- Unit: Transformer Blocks & MoE
- Guide: Ch. 1.1, 2.2, 3.1, 11.1-9, 30.1

5. SFT

Supervised Fine-Tuning, Instruction Templates, Sequence Packing & LoRA Adapters

- Unit: ChatML, Packing & LoRA
- Guide: Ch. 1.8, 3.2, 7.6, 9.4, 30.2

6. Post-Train

Preference Alignment (RLHF/DPO) & Verifiable-Reward Reinforcement Learning (RLVR/GRPO)

- Unit: Pairs, Rollouts & Advantages
- Guide: Ch. 1.8, 3.2, 12.1-8, 30.2

7. Test-Time

Test-Time Compute Scaling, System-2 Chain-of-Thought Reasoning & Tree Search

- Unit: <think> Tokens, PRM & Trees
- Guide: Ch. 13.1-8, 15.1-9, 30.2

8. Context

Context Engineering, Four-Tier Memory, Tool Calling, MCP & Skills

- Unit: Context, Memory & JSON-RPC
- Guide: Ch. 17, 18.2-5, 22, 23

9. Harness

Single-Agent Harness Engineering, Observe-Decide-Act Loops & Stateful Graph Workflows

- Unit: State, Actions & Git Diffs
- Guide: Ch. 16, 19, 20, 27

10. Swarms

Multi-Agent Orchestration, Hierarchical Swarms, A2A Protocols & Shared State

- Unit: A2A Tasks & Swarm State
- Guide: Ch. 1.14, 14, 24, 26, 28, 30.10

CORE SPINE

LLM

Autoregressive Text, Code & Reasoning Language Models

Token-by-Token Language, Symbolic Math, Code Gen & Autonomous Agent Loops

- 1-3: GPU / PyTorch / BPE
- 4-7: MoE / SFT / RLVR / CoT
- 8-10: MCP / Harness^2 / A2A

H100 / B200 · CUDA

HBM3e vs. On-Chip SRAM

FlashAttention-3 Tiling

- CUDA · OpenAI Triton
- FP8 & Microscaling FP4
- NVLink 5 & InfiniBand
- MFU > 45% Utilization
- Prefill vs. Decode BW
- Confidential GPU TEEs

⚠ ANTI-PATTERNS & TRAPS

- × HBM Bandwidth Wall
- × CUDA OOM Crashes
- × Silent SDC Bit-Flips
- × Straggler GPU Stall

uv · PyTorch · vLLM

brew · uv · conda · pip · venv

PyTorch 2.x · JAX · SGLang

- torch.compile & XLA
- PagedAttention KV Cache
- uv.lock Reproducibility
- E2B / Firecracker VM
- Bubblewrap / Seatbelt
- TTFT / TPOT Async SLAs

⚠ ANTI-PATTERNS & TRAPS

- × CUDA ABI & Wheel Hell
- × Unsandboxed Host RCE
- × KV-Cache Preemption
- × pip Breaking flash-attn

FineWeb · Byte-BPE

FineWeb-Edu · Dolma · Stack

Byte-BPE · Tiktoken · Dedup

- Datatrove · NeMo Curator
- Verified Code/Math Synth
- 13-Gram Decontamination
- PII & Secret Scrubbing
- 128k–256k Vocab Size
- Curriculum Annealing

⚠ ANTI-PATTERNS & TRAPS

- × Synthetic Model Collapse
- × Test-Set Contamination
- × Tokenizer Glitch Tokens
- × PII Baked into Weights

Sparse MoE · MLA

Sparse MoE · GQA / MLA

Megatron-LM · TorchTitan

- Llama 4 · Qwen 3 · V3
- Mamba / DeltaNet Hybrid
- RMSNorm, SwiGLU & RoPE
- 3D Parallel (DP/TP/PP/EP)
- Chinchilla Scaling Laws
- Perplexity (BPB) & NIAH

⚠ ANTI-PATTERNS & TRAPS

- × Loss Spikes & NaN Crash
- × MoE Routing Collapse
- × Pipeline Bubble Stall
- × RoPE Length Collapse

ChatML · TRL · LoRA

TRL · Axolotl · Unsloth

LoRA · QLoRA · DoRA PEFT

- Completion Loss Masking
- Multipack + Position IDs
- Tool & Trajectory SFT
- Cold-Start CoT Traces
- MergeKit (SLERP / DARE)
- IFEval Format Audits

⚠ ANTI-PATTERNS & TRAPS

- × Unmasked Prompt Loss
- × EOS/BOS Template Loops
- × Packed Attention Bleed
- × Catastrophic Forgetting

PPO · DPO · GRPO

PPO · DPO · KTO · SimPO

RLVR: GRPO · DAPO · verl

- DeepSeek-R1 · o3 · QwQ
- Code/Math Test Oracles
- OpenRLHF · NeMo Aligner
- KL & Entropy Monitoring
- Deliberative Alignment
- Length-Controlled Win

⚠ ANTI-PATTERNS & TRAPS

- × Reward Hacking & Bloat
- × Policy Entropy Collapse
- × Zero-Variance Advantage
- × Judge Sycophancy Bias

<think> CoT · MCTS

System-2 <think> CoT

Best-of-N · PRM · MCTS

- o3 · Claude 4.6 · R1
- Step-Level Process RM
- Adaptive Budget Forcing
- Exec-Filter Before Vote
- Pass@k vs. Majority@k
- AIME & ARC-AGI-2 Evals

⚠ ANTI-PATTERNS & TRAPS

- × 10x Overthinking Burn
- × PRM Verifier Hacking
- × Self-Correction Regress
- × Unfaithful Post-Hoc CoT

MCP · 4-Tier Mem

Model Context Protocol

4-Tier Memory & SKILL.md

- Letta · Mem0 · Zep
- Episodic & Semantic Mem
- Progressive Skill Load
- Sleep-Time Consolidation
- /compact (<65% Window)
- MCP OAuth 2.1 & LoCoMo

⚠ ANTI-PATTERNS & TRAPS

- × Lost-in-the-Middle Rot
- × 100+ MCP Schema Bloat
- × SKILL.md / MCP Rug-Pull
- × Poisoned Memory State

Claude Code · Loop

Claude Code · Codex · Cursor

LangGraph · Harness^2 Loop

- CLAUDE.md / AGENTS.md
- Observe-Decide-Act Loop
- Typed State & Checkpoints
- Spec->Plan->Test->Verify
- Independent Test Grader
- SWE-bench & pass^k SLA

⚠ ANTI-PATTERNS & TRAPS

- × Deleting Failing Tests
- × Infinite Loop Token Burn
- × God-Node State Bloat
- × IDE Auto-Approve RCE

A2A · OWC · AG-UI

Orchestrator–Worker–Critic

Google A2A · CrewAI · AG2

- Parallel Git Worktrees
- Red-vs-Blue Independent
- CopilotKit · AG-UI
- Agent Gateway & FinOps
- Cryptographic Agent KYA
- /.well-known/agent.json

⚠ ANTI-PATTERNS & TRAPS

- × Swarm Echo Hallucinate
- × $O(N^2)$ Chatty Token Burn
- × Confused Deputy Breach
- × Worktree Merge Conflict

ORGANISM · KNOWLEDGE

RAG

Hybrid Vector, Lexical & Graph Retrieval-Augmented Generation

External Grounding, Dense + BM25 Search, GraphRAG & Citation Verification

- 1-3: SIMD / FAISS / Docling
- 4-7: BGE-M3 / Rerank / HyDE
- 8-10: Qdrant / Agentic RAG

cuVS · SIMD · NVMe

NVIDIA cuVS / RAFT GPU

AVX-512 Distance SIMD

- CAGRA GPU Graph Index
- NVMe Gen5 DiskANN I/O
- Int8 / Binary PQ Codes
- RDMA Index Sharding
- Sub-10ms p99 Vector SLA
- DRAM + SSD Tiered Pool

⚠ ANTI-PATTERNS & TRAPS

- × SSD Random-Read Thrash
- × HNSW Exceeding DRAM
- × PCIe Index Bottleneck
- × NUMA Socket Latency

FAISS · Arrow · PQ

FAISS · HNSWlib · LanceDB

psycopg3 · qdrant-client

- PyArrow Zero-Copy RAM
- BM25 · Tantivy · DuckDB
- Parquet Columnar Store
- Async gRPC / HTTP Pools
- ONNX / TRT Embed Serve
- mmap Vector Segments

⚠ ANTI-PATTERNS & TRAPS

- × Python GIL Chunk Lock
- × Pickle Index RCE Exploit
- × DB Pool Exhaustion
- × Pandas Copy RAM Blowup

Docling · Chunking

Docling · Marker PDF ETL

Contextual & AST Chunking

- Anthropic Chunk Headers
- Table-to-Markdown Parse
- Hard-Negative Mining
- Metadata & RBAC Tags
- CDC Incremental Sync
- Golden QA Eval Sets

⚠ ANTI-PATTERNS & TRAPS

- × Mid-Table Naive Split
- × PDF Column Scramble
- × Stale Deleted Vectors
- × Missing ACL Metadata

BGE-M3 · ColBERTv2

Bi-Encoder InfoNCE Loss

BGE-M3 · NV-Embed · ColBERT

- Matryoshka (MRL) Dims
- SPLADE Sparse Lexical
- Cross-GPU In-Batch Negs
- Bidirectional Encoders
- Multilingual Alignment
- MTEB & NDCG@10 Evals

⚠ ANTI-PATTERNS & TRAPS

- × Embedding Anisotropy
- × False-Negative Batches
- × Domain Acronym Blindness
- × 512-Token Truncation

Rerank · RAFT SFT

Cross-Encoder bge-reranker

Citation-Grounded RAFT SFT

- Domain Embed Fine-Tune
- Distractor Doc Immunity
- Query-Rewrite SFT
- Text-to-SQL / JSON SFT
- Margin-MSE Distillation
- 'I Don't Know' Abstain

⚠ ANTI-PATTERNS & TRAPS

- × Memorizing Stale Facts
- × Believing Distractors
- × Fake [1] Citation Tags
- × Narrow Query Overfit

Search-R1 · DPO

Search-R1 Multi-Hop RLVR

Citation Faithfulness DPO

- Query-Retrieve-Reason RL
- Exact-Match / F1 Oracles
- NLI Entailment Rewards
- Conflict Resolution RL
- Search Cost Penalty RL
- RAGTruth & FaithBench

⚠ ANTI-PATTERNS & TRAPS

- × Redundant Search Loops
- × Cherry-Picking Evidence
- × Over-Abstention Refusal
- × Uncited Extrapolation

RRF · CRAG · HyDE

Hybrid RRF + Reranking

IRCoT · CRAG · Self-RAG

- Query Decomposition
- HyDE Hypothetical Docs
- Top-100 -> 10 Reranker
- Interleaved Search + CoT
- Minimum-Span Extraction
- Dynamic Score Cutoffs

⚠ ANTI-PATTERNS & TRAPS

- × Infinite Rewrite Loops
- × HyDE Hallucination Drift
- × Reranker p99 SLA Blowup
- × Multi-Hop Error Cascade

Qdrant · GraphRAG

Qdrant · Milvus · pgvector

GraphRAG · LightRAG · Neo4j

- REFRAG / LLMLingua Comp
- Dense + BM25 Hybrid DB
- Leiden Graph Communities
- Parent-Doc Expansion
- Tenant RBAC Pre-Filter
- RAG Triad Faithfulness

⚠ ANTI-PATTERNS & TRAPS

- × Indirect Prompt Inject
- × Cross-Tenant Vector Leak
- × Context Stuffing Noise
- × GraphRAG Indexing Cost

Agentic RAG Loop

Agentic RAG Workflows

Plan–Search–Read–Verify

- Adaptive Router Gate
- Sub-Question DAG Merge
- Span-Level Citation Gate
- Source Freshness Check
- Web Fallback on KB Miss
- DeepResearch Benchmark

⚠ ANTI-PATTERNS & TRAPS

- × Rephrase Search Thrash
- × Silent Hallucination
- × Citation Laundering
- × Unbounded Research Lag

Federated RAG MAS

Federated Knowledge Broker

Parallel Deep-Research MAS

- SQL + Vector + Graph Bus
- Map-Reduce Lit Synthesis
- Fact-Checker vs. Writer
- OAuth On-Behalf-Of ACLs
- Shared Citation Board
- Cross-Silo Recall Evals

⚠ ANTI-PATTERNS & TRAPS

- × Circular Citation Loop
- × Cross-Dept ACL Leakage
- × Duplicate Doc Fetching
- × Dropping Minority Facts

ORGANISM · SENSORY

Multimodal

Vision, Audio, Speech, Video & Native Omni-Modal Models

ViT Patch Perception, Native Speech Codecs, Spatio-Temporal Video & Cross-Modal Omni

- 1-3: NVDEC / FFmpeg / Pairs
- 4-7: SigLIP / LLaVA / Visual CoT
- 8-10: Token Merge / LiveKit

NVDEC · ViT · DSP

Hardware NVDEC / NVENC

ViT Patch SRAM Tiling

- Apple M4 Unified RAM
- JPEG/HEVC HW Decode
- 2D/3D DiT Conv Kernels
- GPUDirect Video Ingest
- Sub-200ms Voice DSP
- 4K Frame BW Budgeting

⚠ ANTI-PATTERNS & TRAPS

- × CPU Video Decode Stall
- × 4K Frame PCIe Flood
- × Audio Sample Clock Drift
- × Video Thermal Throttle

FFmpeg · OpenCV

torchvision · torchaudio

LiveKit · whisper.cpp

- FFmpeg · PyAV · Decord
- NVIDIA DALI GPU Loader
- Pillow-SIMD · librosa
- vLLM Multimodal Serve
- Silero VAD Turn Runtime
- WebRTC Jitter Buffers

⚠ ANTI-PATTERNS & TRAPS

- × OpenCV BGR vs RGB Bug
- × FFmpeg Zombie Leaks
- × Audio Buffer Underruns
- × Pillow Bomb DoS Crash

DataComp · OCR

DataComp · OBELICS Pairs

OCR, Chart & Speech Sets

- CLIPScore Aesthetic Gate
- ShareGPT4V Recaptioning
- Scene-Cut Video Clips
- EnCodec / Mimi Tokens
- Face Blur & C2PA Scrub
- Set-of-Mark BBox Labels

⚠ ANTI-PATTERNS & TRAPS

- × Mismatched Alt-Text
- × Low-DPI OCR Misreads
- × Demographic Bias Skew
- × Audio Crosstalk Bleed

SigLIP · Omni · DiT

SigLIP-2 · DINOv2 Vision

Early-Fusion Omni & DiT

- Qwen2.5-VL · InternVL3
- MLP / Q-Former Projector
- 3D Video RoPE Position
- RVQ Audio Codecs (Mimi)
- Flow-Matching Backbones
- Zero-Shot VQA & ASR

⚠ ANTI-PATTERNS & TRAPS

- × Modality Text-IQ Drop
- × Bag-of-Patches Blindness
- × 4K Quadratic ViT Blowup
- × Small-Text Patch Blur

LLaVA · AnyRes SFT

Visual SFT (LLaVA / AnyRes)

GUI BBox & Duplex Voice SFT

- Projector + LLM Stage-2
- High-Res Chart/Doc SFT
- Interleaved Multi-Turn
- Set-of-Mark Visual SFT
- Moshi Full-Duplex SFT
- MMMU & ChartQA Evals

⚠ ANTI-PATTERNS & TRAPS

- × Blind Visual Guessing
- × Projector Detail Loss
- × BBox Coordinate Drift
- × Voice SFT Hurting Math

VLM RLVR · RLAIF-V

Visual Math & OCR RLVR

RLAIF-V & Diffusion DPO

- IoU BBox Reward Oracles
- Chart/Table Verifier RL
- Flow-GRPO Aesthetic RL
- Voice Turn-Taking RLHF
- Visual Safety Alignment
- HallusionBench Audits

⚠ ANTI-PATTERNS & TRAPS

- × Text-Prior Blind Guess
- × Oversaturated Image Hack
- × Benign Image Refusals
- × Voice Barge-In Jitter

Visual CoT · Zoom

Iterative Crop/Zoom CoT

Active ROI Scan & Verify

- o3-Vision Crop Inspect
- OpenCV Code-Exec Zoom
- Video Keyframe Search
- Diffusion CFG Scaling
- Audio Re-Listen Verify
- Multi-Crop Voting

⚠ ANTI-PATTERNS & TRAPS

- × Zooming Irrelevant Bg
- × Blurry Crop Hallucinate
- × High-Res Token Overflow
- × CFG Over-Step Artifacts

ToMe · Video KV

ViT Token Merge (PruMerge)

Streaming Video KV Buffer

- Drop 75% Bg ViT Tokens
- Cross-Modal CLIP Memory
- Audio + Text Dual Index
- MCP Presigned Media URI
- Duplex Barge-In State
- Video-MME Long Context

⚠ ANTI-PATTERNS & TRAPS

- × 10m Video Filling 128k
- × Pruning Tiny OCR Text
- × Base64 JSON-RPC Crash
- × Audio-Video Sync Drift

LiveKit · Pipecat

LiveKit / Pipecat Voice

Render–Inspect–Fix Loop

- VAD + Barge-In Machine
- VLM UI Screenshot Diff
- ComfyUI Gen-Critique
- ControlNet Inpaint Loop
- C2PA Media Provenance
- WER & Turn-Latency SLA

⚠ ANTI-PATTERNS & TRAPS

- × Talking Over User Voice
- × Transcribing Own TTS
- × VLM Breaking CSS Layout
- × >3s Cascaded Voice Lag

Multi-Modal Swarm

Cross-Modal Media Swarm

Director–Video–Audio MAS

- Voice + Tool Subagents
- Character Style Board
- Scene + Foley Sync Gen
- VLM Visual QA Gatekeeper
- Streaming AG-UI Canvas
- Copyright & Brand Gate

⚠ ANTI-PATTERNS & TRAPS

- × Character Identity Drift
- × Voice Promising Failed Op
- × Audio-Video Lip Desync
- × Runaway Render Spend

ORGANISM · ACTION

Embodied

GUI Computer Use, Physics World Models & VLA Robotics

Browser/OS GUI Control, 3D Physics Simulators, World Models & VLA Robot Policies

- 1-3: Edge NPU / Isaac / Teleop
- 4-7: VLA π0 / ACT / Sim2Real
- 8-10: 3DGS / OSWorld / Fleets

Jetson Thor · PhysX

Edge NPU & Sensor ISP

GPU PhysX / Warp Sim

- 1 kHz Real-Time Bus
- LiDAR / IMU Hardware
- EtherCAT / CAN-FD Bus
- INT4/FP8 Edge NPU
- 10,000x Parallel Sim
- Hardware E-Stop Relay

⚠ ANTI-PATTERNS & TRAPS

- × Edge Thermal Throttle
- × OS Jitter Breaking 1kHz
- × CSI Camera Frame Drop
- × Sim Contact Mismatch

ROS 2 · Isaac · CDP

ROS 2 · Gymnasium · MuJoCo

Playwright · LeRobot

- NVIDIA Isaac Lab / Gym
- CycloneDDS Middleware
- Headless Chromium Xvfb
- Open3D · Pinocchio IK
- Async Vectorized Envs
- Deterministic Replays

⚠ ANTI-PATTERNS & TRAPS

- × ROS 2 Multicast Storms
- × Zombie Browser Leaks
- × Euler Gimbal Lock Bug
- × Sim Step Drift Desync

Open-X · GUI Traces

Open X-Embodiment · DROID

Mind2Web · OSWorld · Ego4D

- ALOHA / VR Teleop Demos
- DOM + Screen + Action
- 3D Point-Cloud Align
- Joint-State Normalize
- RLDS / LeRobot Parquet
- Sim Domain Randomize

⚠ ANTI-PATTERNS & TRAPS

- × Teleop Hesitation Noise
- × Screen Resolution Drift
- × Action-Space Mismatch
- × Unlogged Confounders

VLA π0 · V-JEPA 2

VLA (π0 · OpenVLA · RT-2)

World Models (V-JEPA / Genie)

- Diffusion Action Policy
- Octo Multi-Embodiment
- Latent Next-Frame Pred
- 3D Point-Cloud Encoder
- SE(3) Equivariant Pose
- Zero-Shot Sim Transfer

⚠ ANTI-PATTERNS & TRAPS

- × Spurious Causal Confusion
- × World Physics Delusion
- × Mode-Averaging Freeze
- × Camera Viewpoint Fragile

ACT · BC · OS-Atlas

Behavioral Cloning & ACT

GUI Action SFT (OS-Atlas)

- k-Step Action Chunking
- Language-Conditioned BC
- Click / Type / Scroll SFT
- Wrist + 3rd-Person View
- New-Gripper LoRA Tune
- Offline Action MSE Eval

⚠ ANTI-PATTERNS & TRAPS

- × Compounding BC Drift
- × Copycat Action Inertia
- × Lighting Overfitting
- × Static Coordinate Click

Sim2Real · RLVR

OSWorld / WebArena RLVR

Sim-to-Real PPO & HIL-SERL

- DOM / DB State Oracles
- Human-in-Loop SERL RL
- Friction/Mass Randomize
- Residual VLA Policy RL
- Control-Barrier Shield
- Docker Rollout Resets

⚠ ANTI-PATTERNS & TRAPS

- × Simulator Physics Exploit
- × Hardware Damage in RL
- × Reward-Hacking 'Buy' Btn
- × Sparse Credit Stall

MPC · Tree Rollout

Model-Predictive Control

World-Model Tree Search

- CEM / MPPI Sim Rollouts
- VLM Failure Replanning
- Pre-Click Hover Verify
- VoxPoser 3D Value Maps
- Receding-Horizon Exec
- Safety-Filter Critic

⚠ ANTI-PATTERNS & TRAPS

- × Planning Lag > 1kHz Loop
- × 5s Rollout Physics Drift
- × Unverified Destructive Click
- × Obstacle Deadlock Loop

3DGS · DOM AXTree

3D Scene Graphs & 3DGS

DOM & AXTree Pruning

- OctoMap Occupancy Grid
- Set-of-Mark Element IDs
- Proprioceptive Ring Buf
- Kalman 6-DoF Object Track
- MCP Robot/Browser Bus
- Spatial Consistency QA

⚠ ANTI-PATTERNS & TRAPS

- × 50k-Node Raw DOM Flood
- × Dynamic Obstacle Ghosts
- × Out-of-Sight Forgetting
- × Odometry Frame Drift

Computer Use Loop

Operator / Claude CU Loop

System-2 VLM + System-1 VLA

- Screen -> Action -> Verify
- Pop-Up & Auth Recovery
- 2Hz Planner + 50Hz Motor
- Force/Torque Reflex Stop
- Checkout/Delete Gate
- OSWorld / WebArena Eval

⚠ ANTI-PATTERNS & TRAPS

- × Fake Pop-Up Injection
- × Infinite 'Pay' Clicks
- × UI Lag Misaligned Click
- × Ignoring Gripper Slip

Fleet MAPF Swarm

Multi-Robot Fleet Control

Digital-Twin + Physical MAS

- Conflict-Free MAPF Path
- Drone + Rover + Arm MAS
- V2X Spatial Perception
- Cloud Plan + Edge Reflex
- Human Teleop Handoff
- Fleet Deadlock Evals

⚠ ANTI-PATTERNS & TRAPS

- × Corridor Traffic Gridlock
- × Faulty Sensor Map Poison
- × Wi-Fi Drop Fleet Split
- × Dual-Arm Grasp Collision